MIL-Y-11241D(CmlC)
29 December 1961
SUPERSEDING
MIL-Y-11241C(CmlC)
28 March 1960

MILITARY SPECIFICATION

YEAST AUTOLYSATE, PASTE, TECHNICAL

1. SCOPE

1.1 This specification covers a paste type of water soluble yeast autolysate for use as a constituent nutrient in biological media.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-760 - Drums and Pails, Metal (5 and 16.64 Gallon)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASSOCIATION OF OFFICIAL AGRICULTURAL CHEMISTS

Official Methods of Analysis, Eighth Edition, 1955.

(Application for copies should be addressed to the Association of Official Agricultural Chemists, P. O. Box 540, Benjamin Franklin Station, Washington 4, D. C.)

3. REQUIREMENTS

3.1 Product. The product shall be an unadulterated autolysate made from a culture of Saccharomyces cerevisiae of good commercial quality.

3.1.1 Autolysate. The autolysate shall be a concentrate of the water soluble portion of the yeast cells. No chemicals, other than sodium chloride and the acids or bases used for preliminary pH adjustment, shall be used for the production of the autolysate. (The autolysate shall not be a reconstituted powder.)

3.2 Odor. The autolysate shall have a characteristic but not putrescent odor.

3.3 Nutritional value. The product covered by this specification shall supply sufficient nutritional value to support adequate growth of various micro-organisms (see 6.2) when tested as specified in 4.6.7.

3.4 Pre-award sample. At no cost to the Government, each bidder shall forward a sample of the first lot of autolysate intended for shipment under the contract or order to the destination indicated by the contracting officer. The Government will test the samples to determine compliance with the requirements of this specification.

3.5 Chemical and physical properties. The chemical and physical properties of the autolysate shall conform to the requirements given in table I when tested as specified in the test paragraphs referenced therein.

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Table I - Chemical and Physical Properties

Property	Requirements*	Test Paragraphs
Total solids**	75 ± 3%	4.6.1
Total nitrogen**	5.5% (min.)	4.6.2
Amino nitrogen**	Not less than 35% of the total nitrogen content	4.6.3
Ash**	28.5% (max.)	4.6.4
Sodium chloride**	16.5% to 19.5%	4.6.5
pH (in solution)	5.0 to 6.2	4.6.6

*Percentage requirements are by weight.
**On an "as is" basis.

3.6 Acceptance lots. The lot or lots submitted to the Government for final acceptance shall be the lot or lots from which the acceptance samples (see 4.4.2) were selected and approved by the Government as complying with the requirements of this specification.

3.7 Storage conditions. During the period between the taking of the acceptance samples and the time of shipment, the lot or lots shall not be subjected to any known condition which may have adverse affects on the nutritional value of the autolysate.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Supplier's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government responsibility. The Government will perform the tests specified in 4.6.7.

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4.2 Special provisions.

4.2.1 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Supplemental evidence. The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, 3.6, 3.7, and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of the autolysate produced in the final concentrating operation by one manufacturer in not more than 24 consecutive hours, under essentially the same manufacturing conditions, with no recharging of the final concentrator feed tank, providing the process is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.4 Sampling.

4.4.1 Pre-award sample. A 1000 gram (g) representative sample of the autolysate shall be placed in a clean, dry container labeled in accordance with Standard MIL-STD-129.

4.4.2 Acceptance sample.

4.4.2.1 For nondestructive examination. Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.4.2.2 For test. Sample containers shall be taken at random from each lot as follows:

Lot size	Sample size
1	1
2- 275	2
276- 545	3
546- 900	4
901-1345	5
1346-1875	6
1876-2500	7

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Sample size for lots exceeding 2500 containers shall be calculated using the following equation:

$$n = 0.15\sqrt{N}$$

where: n = sample size.
N = lot size.

A 1000-g specimen shall be removed from each container in the sample and placed in a clean, dry container labeled to identify the lot and container from which it was taken. Each specimen shall be separately tested as specified in 4.6.

4.5 Examination. Sample containers of the autolysate shall be examined for compliance with level A preparation for delivery requirements in accordance with the classification of defects and with Standard MIL-STD-105.

4.5.1 Classification of defects.

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4.5.1.1 Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Pail leaking
102	Pail incorrect
103	Marking missing or incorrect

4.6 Tests. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Where applicable, blank determinations shall be conducted in parallel with the tests using the same quantities of the reagents, and correction applied when significant. Unless otherwise specified, all weighings shall be made to an accuracy of 1 milligram. Tests shall be conducted as follows:

4.6.1 Total solids. Weigh 1 g of the specimen in a previously tared ground-glass stoppered weighing bottle, spreading the specimen evenly on the bottom of the bottle as it is introduced. Dry the specimen to a constant weight in a vacuum oven at a temperature of 80° to 85°C and an absolute pressure of 130 ± 5 millimeters of mercury. Calculate the percentage of total solids using the formula:

$$\text{percent total solids} = \frac{\text{weight of dried specimen}}{\text{initial weight of specimen}} \times 100$$

4.6.2 Total nitrogen. Weigh 1 g of the specimen in a previously tared 50-milliliter (ml) volumetric flask. Fill the flask to the mark with water and thoroughly mix the contents. Pipette 10.0 ml of this dilution into a Kjeldahl flask. (Save the remainder for the test specified in 4.6.3.) Determine the total nitrogen in accordance with paragraph 2.23 of Official Methods of Analysis of the Association of Official Agricultural Chemists.

4.6.3 Amino nitrogen. Pipette 5.0 ml of the dilution prepared for the test specified in 4.6.2 into a Van Slyke apparatus. Determine the amino nitrogen in accordance with paragraphs 23.20 and 23.21 of Official Methods of Analysis of the Association of Official Agricultural Chemists. Calculate the percentage of amino nitrogen on the basis of the total nitrogen (see 4.6.2).

4.6.4 Ash. Determine the ash content of the specimen in accordance with paragraphs 23.34 and 29.12 of Official Methods of Analysis of the Association of Official Agricultural Chemists.

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4.6.5 Sodium chloride. Weigh 1.1 g of the specimen in a previously tared porcelain or nickel crucible. Add 1 ml of a concentrated solution of sodium hydroxide to the specimen and mix with a clean glass stirring rod. Wipe the rod with a wet filter paper and place the paper in the crucible. Heat the crucible in an oven at a temperature of 102° ± 2°C until the water is expelled, and then place it in a muffle furnace at 525°C until a white ash is obtained. Remove the crucible from the furnace and allow it to cool to room temperature. Dissolve the contents of the crucible in hot water and filter the solution into a 200-ml volumetric flask. Rinse the crucible and the filter paper several times with hot water and add the rinse water to the filtrate. Acidify the filtrate with 2 ml of concentrated nitric acid and add 50.0 ml of 0.1N silver nitrate. Mix the contents of the flask, allow to cool, dilute the mixture to the mark, and mix thoroughly. Pipette 100.0 ml of the contents of the flask into a 250-ml Erlenmeyer flask, add 0.3 g of solid ammonium ferrosulfate, and titrate the mixture with 0.1N ammonium thiocyanate. Calculate the percentage of sodium chloride as follows:

$$\text{Percent sodium chloride} = \frac{11.69\ (AB-CD)}{W}$$

where: A = ml silver nitrate divided by 2.
B = normality of silver nitrate.
C = ml ammonium thiocyanate.
D = normality of ammonium thiocyanate.
W = weight of specimen in g.

4.6.6 pH (in solution). Dissolve 2.00 g of the specimen in 50 ml of recently boiled water in a 100-ml volumetric flask, dilute to the mark with additional carbon dioxide free water and thoroughly mix the solution. Determine the pH of the solution at 25°C, using a glass electrode - saturated calomel electrode - pH meter system (see 6.4) in accordance with the instructions on the meter.

4.6.7 Nutritional value. The Government will combine the autolysate with known standard materials to form a medium and determine its nutritional value by means of the growth rate of various micro-organisms.

4.7 Acceptance/rejection criteria. If any specimen of the autolysate fails to meet the requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. The autolysate shall be packed in a 5-gallon pail conforming to type II, class 1, of Specification PPP-D-760.

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5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) That pre-award samples will be required.

6.2 Nutritional value. Due to unknown factors affecting the growth of various micro-organisms and the possible danger in handling these organisms, the Government will conduct all tests on samples to establish the nutritional value of the product. It is estimated that within 30 days after receipt of the pre-award samples or acceptance samples the prospective bidders or supplier will be notified of the acceptance or rejection of the product on the basis of nutritional value.

6.3 Batch. For the purpose of this specification, a batch is defined as that quantity of material produced from each charge of the final concentrating vessel.

6.4 pH meters and systems. pH meters and systems complying with the requirements of ASTM Designation E70 have been found to be satisfactory for pH determinations.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, and other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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